Hartree-Fock-Bogoliubov theory

W. Ryssens

Final class

- Today is (likely) the final class
- Q& A session on the **7th of May**
- Questions sent beforehand will improve it.

About the exam

The exam is

- Open book - bring whatever you want.
- Problem solving - be prepared to apply what you learned.

Additional reading

- Quantum theory of finite systems
  J.-P. Blaizot and G. Ripka
  MIT Press (1986).
- The nuclear many-body problem
  P. Ring and P. Schuck
  Springer-Verlag (1980).

Feedback

Please fill in your student evaluations.

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
    - Parity
    - Angular momentum
    - Gauge rotation
    - Translation

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
    - Parity
    - Angular momentum
    - Gauge rotation
    - Translation
4. Symmetric Slater determinants: E, ρ and h.

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
    - Parity
    - Angular momentum
    - Gauge rotation
    - Translation
4. Symmetric Slater determinants: E, ρ and h.
5. Symmetries as variational restrictions

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
   - Parity
   - Angular momentum
   - Gauge rotation
   - Translation
4. Symmetric Slater determinants: E, ρ and h.
5. Symmetries as variational restrictions
6. Symmetry-broken Slater determinants

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
   - Parity
   - Angular momentum
   - Gauge rotation
   - Translation
4. Symmetric Slater determinants: E, ρ and h.
5. Symmetries as variational restrictions
6. Symmetry-broken Slater determinants
7. Order parameters and correlation energy

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
    - Parity
    - Angular momentum
    - Gauge rotation
    - Translation
4. Symmetric Slater determinants: E, ρ and h.
5. Symmetries as variational restrictions
6. Symmetry-broken Slater determinants
7. Order parameters and correlation energy
8. Excitation patterns

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
    - Parity
    - Angular momentum
    - Gauge rotation
    - Translation
4. Symmetric Slater determinants: E, ρ and h.
5. Symmetries as variational restrictions
6. Symmetry-broken Slater determinants
7. Order parameters and correlation energy
8. Excitation patterns
9. Single-particle quantum numbers

Summary of last time

We discussed

1. symmetries of the nuclear Hamiltonian $\leftrightarrow$ unitary operators U
2. the appearance of quantum numbers
3. Examples
   - Parity
   - Angular momentum
   - Gauge rotation
   - Translation
4. Symmetric Slater determinants: E, ρ and h.
5. Symmetries as variational restrictions
6. Symmetry-broken Slater determinants
7. Order parameters and correlation energy
8. Excitation patterns
9. Single-particle quantum numbers
10. Symmetry restoration through projection

The Hartree-Fock-Bogoliubov ansatz

Single-particle basis $\{|\alpha\rangle\}$ with associated

- annihilation/creation operators such that $\{a_\alpha, a^\dagger_\beta\} = \delta_{\alpha\beta}$.
- single-particle wavefunctions $\psi_\alpha(\vec{r}\sigma q) \equiv \langle \vec{r}\sigma q|\alpha\rangle$

The Hartree-Fock-Bogoliubov ansatz

Single-particle basis $\{|\alpha\rangle\}$ with associated

- annihilation/creation operators such that $\{a_\alpha, a^\dagger_\beta\} = \delta_{\alpha\beta}$.
- single-particle wavefunctions $\psi_\alpha(\vec{r}\sigma q) \equiv \langle \vec{r}\sigma q|\alpha\rangle$

Slater determinant

$$|\Psi\rangle \equiv \prod_{i=1}^{N} b^\dagger_i |0\rangle$$

with $b^\dagger_\mu \equiv \sum_\alpha U_{\alpha\mu} a^\dagger_\alpha$

- is a vacuum for

$$b^\dagger_\mu |\Psi\rangle = 0 \text{ for } \mu \in [1, N]$$

$$b_\mu |\Psi\rangle = 0 \text{ for } \mu \in [N+1, \infty[$$

- has definite particle number

$$N|\Psi\rangle = N_0|\Psi\rangle$$

The Hartree-Fock-Bogoliubov ansatz

Single-particle basis $\{|\alpha\rangle\}$ with associated

- annihilation/creation operators such that $\{a_\alpha, a^\dagger_\beta\} = \delta_{\alpha\beta}$.
- single-particle wavefunctions $\psi_\alpha(\vec{r}\sigma q) \equiv \langle \vec{r}\sigma q|\alpha\rangle$

Slater determinant

$$|\Psi\rangle \equiv \prod_{i=1}^{N} b^\dagger_i |0\rangle$$

with $b^\dagger_\mu \equiv \sum_\alpha U_{\alpha\mu} a^\dagger_\alpha$

- is a vacuum for

$$b^\dagger_\mu|\Psi\rangle = 0 \text{ for } \mu \in [1, N]$$

$$b_\mu|\Psi\rangle = 0 \text{ for } \mu \in [N+1, \infty[$$

- has definite particle number

$$N|\Psi\rangle = N_0|\Psi\rangle$$

Bogoliubov state

See next slide.

$$|\Psi\rangle \equiv \underbrace{\mathcal{N}^{-1/2}}_{\text{Normalisation}} \prod_{\mu=1}^{M} \beta_\mu |0\rangle$$

with $\beta_\mu \equiv \sum_\alpha U^*_{\alpha\mu} a_\alpha + V^*_{\alpha\mu} a^\dagger_\alpha$

- is a vacuum for

$$\beta_\mu|\Psi\rangle = 0 \text{ for all } \mu$$

- ... broken particle number

$$N|\Psi\rangle \neq N_0|\Psi\rangle$$

Bogoliubov vacuum construction

$$|\Psi\rangle \equiv \mathcal{N}^{-1/2} \prod_{\mu=1}^{M} \beta_\mu |0\rangle \, .$$

- If the sum runs over the entire s.p. basis, $|\Psi\rangle$ is likely to vanish.

Bogoliubov vacuum construction

$$|\Psi\rangle \equiv \mathcal{N}^{-1/2} \boxed{\prod_{\mu=1}^{M}} \beta_\mu |0\rangle \,.$$

- If the sum runs over the entire s.p. basis, $|\Psi\rangle$ is likely to vanish.
- "Pure" destruction operators should be banned
  i.e. β_ν is not in the product if $\beta_\nu = a_\nu$.

Bogoliubov vacuum construction

$$|\Psi\rangle \equiv \mathcal{N}^{-1/2} \prod_{\mu=1}^{M} \beta_\mu |0\rangle \,.$$

- If the sum runs over the entire s.p. basis, $|\Psi\rangle$ is likely to vanish.
- "Pure" destruction operators should be banned
  i.e. β_ν is not in the product if $\beta_\nu = a_\nu$.
- The crucial property is that this is a vacuum

$$\beta_\mu |\Psi\rangle = 0 \, \forall \mu$$

which will allow us to bypass any explicit construction of $|\Psi\rangle$.

Bogoliubov vacuum construction

$$|\Psi\rangle \equiv \mathcal{N}^{-1/2} \prod_{\mu=1}^{M} \beta_\mu |0\rangle \, .$$

- If the sum runs over the entire s.p. basis, $|\Psi\rangle$ is likely to vanish.
- "Pure" destruction operators should be banned
  i.e. β_ν is not in the product if $\beta_\nu = a_\nu$.
- The crucial property is that this is a vacuum

  $$\beta_\mu |\Psi\rangle = 0 \, \forall \mu$$

  which will allow us to bypass any explicit construction of $|\Psi\rangle$.
- *M* can be **even** or **odd**.

Bogoliubov vacuum construction

$$|\Psi\rangle \equiv \mathcal{N}^{-1/2} \boxed{\prod_{\mu=1}^{M}} \beta_\mu |0\rangle \, .$$

- If the sum runs over the entire s.p. basis, $|\Psi\rangle$ is likely to vanish.
- "Pure" destruction operators should be banned
  i.e. β_ν is not in the product if $\beta_\nu = a_\nu$.
- The crucial property is that this is a vacuum

$$\beta_\mu |\Psi\rangle = 0 \, \forall \mu$$

  which will allow us to bypass any explicit construction of $|\Psi\rangle$.
- M can be **even** or **odd**.
- $|\Psi\rangle$ acquires only phase when mixing qps unitarily $\gamma_\mu = \sum_\alpha U_{a\mu} \beta_\alpha$

Bogoliubov vacuum construction

$$|\Psi\rangle \equiv \mathcal{N}^{-1/2} \prod_{\mu=1}^{M} \beta_\mu |0\rangle \, .$$

- If the sum runs over the entire s.p. basis, $|\Psi\rangle$ is likely to vanish.
- "Pure" destruction operators should be banned
  i.e. β_ν is not in the product if $\beta_\nu = a_\nu$.
- The crucial property is that this is a vacuum

$$\beta_\mu |\Psi\rangle = 0 \, \forall \mu$$

  which will allow us to bypass any explicit construction of $|\Psi\rangle$.
- M can be **even** or **odd**.
- $|\Psi\rangle$ acquires only phase when mixing qps unitarily $\gamma_\mu = \sum_\alpha U_{a\mu} \beta_\alpha$

  Prove this in analogy to the HF case.

Transformations

Single-particle basis $\{|\alpha\rangle\}$ with associated

- annihilation/creation operators such that $\{a_\alpha, a^\dagger_\beta\} = \delta_{\alpha\beta}$.
- single-particle wavefunctions $\psi_\alpha(\vec{r}\sigma q) \equiv \langle \vec{r}\sigma q|\alpha\rangle$

Slater determinant

$$\begin{pmatrix} b \\ b^\dagger \end{pmatrix} = \begin{pmatrix} U^\dagger & 0 \\ 0 & U \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}$$

- Unitarity implies $U^\dagger U = 1$

Transformations

Single-particle basis $\{|\alpha\rangle\}$ with associated

- annihilation/creation operators such that $\{a_\alpha, a^\dagger_\beta\} = \delta_{\alpha\beta}$.
- single-particle wavefunctions $\psi_\alpha(\vec{r}\sigma q) \equiv \langle \vec{r}\sigma q|\alpha\rangle$

Slater determinant

$$\begin{pmatrix} b \\ b^\dagger \end{pmatrix} = \begin{pmatrix} U^\dagger & 0 \\ 0 & U \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}$$

- Unitarity implies $U^\dagger U = 1$

Bogoliubov state

$$\begin{pmatrix} \beta \\ \beta^\dagger \end{pmatrix} = \mathcal{W} \begin{pmatrix} a \\ a^\dagger \end{pmatrix} = \boxed{\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix}} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}$$

- Unitarity implies $\mathcal{W}^\dagger \mathcal{W} = 1$

  which means:

$$U^\dagger U + V^\dagger V = 1$$
$$U^\dagger V + V^\dagger U = 0$$

Contractions with respect to a Bogoliubov vacuum

We take the Bogoliubov state $|\Psi\rangle \equiv \prod_\mu \beta_\mu|0\rangle$ as reference vacuum with the β_μ linked to a_μ through a Bogoliubov transformation.

For quasiparticle operators:

$$\overbracket{\beta^\dagger_\mu\beta_\nu} = \langle\Psi|\beta^\dagger_\mu\beta_\nu|\Psi\rangle = 0$$
$$\overbracket{\beta_\mu\beta_\nu} = \langle\Psi|\beta_\mu\beta_\nu|\Psi\rangle = 0$$
$$\overbracket{\beta^\dagger_\mu\beta^\dagger_\nu} = \langle\Psi|\beta^\dagger_\mu\beta^\dagger_\nu|\Psi\rangle = 0$$
$$\overbracket{\beta_\mu\beta^\dagger_\nu} = \langle\Psi|\beta_\mu\beta^\dagger_\nu|\Psi\rangle = \delta_{\mu\nu}$$

For particle operators

$$\overbracket{a^\dagger_\mu a_\nu} = \langle\Psi|a^\dagger_\mu a_\nu|\Psi\rangle = + \left[V^* V^T\right]_{\nu\mu}$$
$$\overbracket{a_\mu a_\nu} = \langle\Psi|a_\mu a_\nu|\Psi\rangle = + \left[V^* U^T\right]_{\nu\mu}$$
$$\overbracket{a^\dagger_\mu a^\dagger_\nu} = \langle\Psi|a^\dagger_\mu a^\dagger_\nu|\Psi\rangle = - \left[V U^\dagger\right]_{\nu\mu}$$
$$\overbracket{a_\mu a^\dagger_\nu} = \langle\Psi|a_\mu a^\dagger_\nu|\Psi\rangle = \delta_{\mu\nu} - \left[V V^\dagger\right]_{\nu\mu}$$

Try this at home.

Density matrices

Hermitian normal density

$$\rho_{\alpha\beta} \equiv \overbrace{a^\dagger_\beta a}_\alpha = [V^* V^T]_{\alpha\beta} = \rho^*_{\beta\alpha}$$

Slater determinant

- Fixed $N \leftrightarrow \kappa \equiv 0$
- ρ is idempotent $\rho^2 = \rho$
- In basis $\{b^\dagger_\mu\}$ diagonalising ρ

$$\rho = \begin{pmatrix} 1 & 0 & \dots & 0 & \dots \\ 0 & \ddots & & & \\ \vdots & & 1 & & \\ 0 & & & 0 & \\ \vdots & & & & \ddots \end{pmatrix}$$

with s.p. occupations $\rho_{\mu\mu} = 0/1$.

Density matrices

Hermitian normal density

$$\rho_{\alpha\beta} \equiv \overline{a^\dagger_\beta a_\alpha} = [V^* V^T]_{\alpha\beta} = \rho^*_{\beta\alpha}$$

Antisymmetric anomalous density

$$\kappa_{\alpha\beta} \equiv \overline{a_\beta a_\alpha} = [V^* U^T]_{\alpha\beta} = -\kappa_{\beta\alpha}$$

Slater determinant

- Fixed $N \leftrightarrow \kappa \equiv 0$
- ρ is idempotent $\rho^2 = \rho$
- In basis $\{b^\dagger_\mu\}$ diagonalising ρ

$$\rho = \begin{pmatrix} 1 & 0 & \dots & 0 & \dots \\ 0 & \ddots & & & \\ \vdots & & 1 & & \\ 0 & & & 0 & \\ \vdots & & & & \ddots \end{pmatrix}$$

with s.p. occupations $\rho_{\mu\mu} = 0/1$.

Bogoliubov state

- $\kappa \neq 0$
- ρ is **not** idempotent $\rho^2 \neq \rho$
- In basis $\{b^\dagger_\mu\}$ diagonalising ρ

$$\rho = \begin{pmatrix} v_1^2 & 0 & \dots & 0 & \dots \\ 0 & \ddots & & & \\ \vdots & & v_N^2 & & \\ 0 & & & v_{N+1}^2 & \\ \vdots & & & & \ddots \end{pmatrix}$$

with s.p. occupations $0 \leq v_\mu^2 \leq 1$.

The generalized density matrix

$$\boxed{\mathcal{R} \equiv \begin{pmatrix} \rho & \kappa \\ -\kappa^* & 1-\rho^* \end{pmatrix}}$$

- Trivial in the quasiparticle basis, i.e.

$$\mathcal{W}^\dagger \mathcal{R} \mathcal{W} = \begin{pmatrix} \langle \beta^\dagger \beta \rangle & \langle \beta \beta \rangle \\ \langle \beta^\dagger \beta^\dagger \rangle & \langle \beta \beta^\dagger \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

The generalized density matrix

$$\boxed{\mathcal{R} \equiv \begin{pmatrix} \rho & \kappa \\ -\kappa^* & 1-\rho^* \end{pmatrix}}$$

- Trivial in the quasiparticle basis, i.e.

$$\mathcal{W}^\dagger \mathcal{R} \mathcal{W} = \begin{pmatrix} \langle \beta^\dagger \beta \rangle & \langle \beta\beta \rangle \\ \langle \beta^\dagger \beta^\dagger \rangle & \langle \beta \beta^\dagger \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

- ... which implies idempotency

$$\mathcal{R}^2 = \mathcal{R}$$

The generalized density matrix

$$\boxed{\mathcal{R} \equiv \begin{pmatrix} \rho & \kappa \\ -\kappa^* & 1-\rho^* \end{pmatrix}}$$

- Trivial in the quasiparticle basis, i.e.

$$\mathcal{W}^\dagger \mathcal{R} \mathcal{W} = \begin{pmatrix} \langle \beta^\dagger \beta \rangle & \langle \beta \beta \rangle \\ \langle \beta^\dagger \beta^\dagger \rangle & \langle \beta \beta^\dagger \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

- ... which implies idempotency

$$\mathcal{R}^2 = \mathcal{R}$$

- There is again a bijection (up to a phase):
  Bogoliubov vacuum $\leftrightarrow$ the generalized density matrix

$$|\Psi\rangle \leftrightarrow \mathcal{R}$$

Expression for the energy

With Wicks theorem, you can check that

$$E^{(\mathrm{HFB})} = \sum_{\alpha\beta} t_{\alpha\beta}\rho_{\beta\alpha} + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\rho_{\gamma\alpha}\rho_{\delta\beta} + \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\kappa^*_{\alpha\beta}\kappa_{\gamma\delta} = E[\rho,\kappa,\kappa^*]\,.$$

Expression for the energy

With Wicks theorem, you can check that

$$E^{(\mathrm{HFB})} = \sum_{\alpha\beta} t_{\alpha\beta}\rho_{\beta\alpha} + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\rho_{\gamma\alpha}\rho_{\delta\beta} + \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\kappa^*_{\alpha\beta}\kappa_{\gamma\delta} = E[\rho, \kappa, \kappa^*] \,.$$

- Compared to Hartree-Fock we have
  1. different ρ
  2. new correlation term dependent on κ, κ^*

Expression for the energy

With Wicks theorem, you can check that

$$E^{(\mathrm{HFB})} = \sum_{\alpha\beta} t_{\alpha\beta}\rho_{\beta\alpha} + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\rho_{\gamma\alpha}\rho_{\delta\beta} + \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\kappa^*_{\alpha\beta}\kappa_{\gamma\delta} = E[\rho, \kappa, \kappa^*] \,.$$

- Compared to Hartree-Fock we have
  1. different ρ
  2. new correlation term dependent on κ, κ^*
- The energy is a functional of ρ, κ and κ^*, subject to
  1. $\rho_{ij} = +\rho^*_{ji}$
  2. $\kappa_{ij} = -\kappa_{ji}$

Gauge rotations

$U = e^{i\alpha N}$ is a symmetry operator $\forall \alpha$.

$$e^{i\alpha N} a^\dagger_\mu e^{-i\alpha N} = e^{+i\alpha} a^\dagger_\mu \, , \qquad e^{i\alpha N} a_\mu e^{-i\alpha N} = e^{-i\alpha} a_\mu \, .$$

Gauge rotations

$U = e^{i\alpha N}$ is a symmetry operator $\forall \alpha$.

$$e^{i\alpha N} a^\dagger_\mu e^{-i\alpha N} = e^{+i\alpha} a^\dagger_\mu \, , \qquad e^{i\alpha N} a_\mu e^{-i\alpha N} = e^{-i\alpha} a_\mu \, .$$

$$e^{i\alpha N} \beta^\dagger_\mu e^{-i\alpha N} \neq e^{+i\alpha} \beta^\dagger_\mu \, , \qquad e^{i\alpha N} \beta_\mu e^{-i\alpha N} \neq e^{-i\alpha} \beta_\mu \, .$$

Gauge rotations

$U = e^{i\alpha N}$ is a symmetry operator $\forall \alpha$.

$$e^{i\alpha N} a^\dagger_\mu e^{-i\alpha N} = e^{+i\alpha} a^\dagger_\mu \,, \qquad e^{i\alpha N} a_\mu e^{-i\alpha N} = e^{-i\alpha} a_\mu \,.$$

$$e^{i\alpha N} \beta^\dagger_\mu e^{-i\alpha N} \neq e^{+i\alpha} \beta^\dagger_\mu \,, \qquad e^{i\alpha N} \beta_\mu e^{-i\alpha N} \neq e^{-i\alpha} \beta_\mu \,.$$

A Bogoliubov state breaks particle number (= gauge rotation) symmetry.

$$|\Psi\rangle \to e^{i\alpha N} |\Psi\rangle$$

$$\begin{pmatrix} \rho & \kappa \\ -\kappa^* & 1-\rho^* \end{pmatrix} \to \begin{pmatrix} \rho & \kappa e^{2i\alpha} \\ -\kappa^* e^{-2i\alpha} & 1-\rho^* \end{pmatrix}$$

$$E[\rho, \kappa, \kappa^*] \to E[\rho, \kappa e^{2i\alpha}, \kappa^* e^{-2i\alpha}]$$

$$= E[\rho, \kappa, \kappa^*]$$

Remember: expressions for one species, but generalisable to two species!

The number parity operator

Except when $\alpha = \pi$, then $e^{i\alpha} = e^{-i\alpha} = -1$ and

$$e^{i\pi N}\beta^\dagger_\mu e^{-i\pi N} = -\beta^\dagger_\mu\,, \qquad e^{i\pi N}\beta_\mu e^{-i\pi N} = -\beta_\mu$$

The number parity operator

Except when $\alpha = \pi$, then $e^{i\alpha} = e^{-i\alpha} = -1$ and

$$e^{i\pi N}\beta^{\dagger}_{\mu}e^{-i\pi N} = -\beta^{\dagger}_{\mu}\,, \qquad e^{i\pi N}\beta_{\mu}e^{-i\pi N} = -\beta_{\mu}$$

$e^{i\pi N}$ = Number parity operator

The number parity operator

Except when $\alpha = \pi$, then $e^{i\alpha} = e^{-i\alpha} = -1$ and

$$e^{i\pi N} \beta^\dagger_\mu e^{-i\pi N} = -\beta^\dagger_\mu \,, \qquad e^{i\pi N} \beta_\mu e^{-i\pi N} = -\beta_\mu$$

$e^{i\pi N}$ = Number parity operator

- This operator has eigenvalues ± 1. Show this.

The number parity operator

Except when $\alpha = \pi$, then $e^{i\alpha} = e^{-i\alpha} = -1$ and

$$e^{i\pi N}\beta^\dagger_\mu e^{-i\pi N} = -\beta^\dagger_\mu , \qquad e^{i\pi N}\beta_\mu e^{-i\pi N} = -\beta_\mu$$

$e^{i\pi N}$ = Number parity operator

- This operator has eigenvalues ± 1. Show this.
- Eigenstates of N are eigenstates of $e^{i\pi N}$

$$e^{i\pi \hat{N}}|N\rangle = (-1)^{N_0}|N_0\rangle .$$

The number parity operator

Except when $\alpha = \pi$, then $e^{i\alpha} = e^{-i\alpha} = -1$ and

$$e^{i\pi N}\beta^\dagger_\mu e^{-i\pi N} = -\beta^\dagger_\mu\,, \qquad e^{i\pi N}\beta_\mu e^{-i\pi N} = -\beta_\mu$$

$e^{i\pi N}$ = Number parity operator

- This operator has eigenvalues ± 1. Show this.
- Eigenstates of N are eigenstates of $e^{i\pi N}$

$$e^{i\pi\hat{N}}|N\rangle = (-1)^{N_0}|N_0\rangle\,.$$

- Bogoliubov states have good number parity

$$e^{i\pi N}|\Psi\rangle = e^{i\pi N}\prod_{\mu=1}^{M}\beta_\mu|0\rangle = (-1)^M|\Psi\rangle\,.$$

The number parity operator

Except when $\alpha = \pi$, then $e^{i\alpha} = e^{-i\alpha} = -1$ and

$$e^{i\pi N}\beta^\dagger_\mu e^{-i\pi N} = -\beta^\dagger_\mu \,, \qquad e^{i\pi N}\beta_\mu e^{-i\pi N} = -\beta_\mu$$

$e^{i\pi N}$ = Number parity operator

- This operator has eigenvalues ± 1. Show this.
- Eigenstates of N are eigenstates of $e^{i\pi N}$

$$e^{i\pi \hat{N}}|N\rangle = (-1)^{N_0}|N_0\rangle \,.$$

- Bogoliubov states have good number parity

$$e^{i\pi N}|\Psi\rangle = e^{i\pi N}\prod_{\mu=1}^{M}\beta_\mu|0\rangle = (-1)^M|\Psi\rangle \,.$$

- Breaking number parity $\leftrightarrow$ statistical mixtures at finite T

Two types of Bogoliubov states

Schematically:

$$\boxed{|\Psi^{\text{even}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha\rangle \,, \qquad \boxed{|\Psi^{\text{odd}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha + 1\rangle \,.$$

Number parity $+1$

Number parity -1

Two types of Bogoliubov states

Schematically:

$$\boxed{|\Psi^{\text{even}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha\rangle \,, \qquad \boxed{|\Psi^{\text{odd}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha + 1\rangle \,.$$

Number parity $+1$

Number parity -1

- $|\Psi^{\text{even}}\rangle$ can be used to model nuclei with even particle number.

Two types of Bogoliubov states

Schematically:

$$\boxed{|\Psi^{\text{even}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha\rangle \,, \qquad \boxed{|\Psi^{\text{odd}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha + 1\rangle \,.$$

Number parity $+1$

Number parity -1

- $|\Psi^{\text{even}}\rangle$ can be used to model nuclei with even particle number.
- $|\Psi^{\text{odd}}\rangle$ can be used to model nuclei with odd particle number.

Two types of Bogoliubov states

Schematically:

$$\boxed{|\Psi^{\text{even}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha\rangle \,, \qquad \boxed{|\Psi^{\text{odd}}\rangle} = \sum_{\alpha=0}^{\infty} c_\alpha |N = 2\alpha + 1\rangle \,.$$

Number parity $+1$

Number parity -1

- $|\Psi^{\text{even}}\rangle$ can be used to model nuclei with even particle number.
- $|\Psi^{\text{odd}}\rangle$ can be used to model nuclei with odd particle number.
- Average particle number can take **ARBITRARY REAL** values.

$\langle\Psi^{\text{even}}|N|\Psi^{\text{even}}\rangle$ can be odd.

$\langle\Psi^{\text{odd}}|N|\Psi^{\text{odd}}\rangle$ can be even.

$\langle\Psi|N|\Psi\rangle$ can be non-integer.

Summary so far

- Particle number broken: $|\Psi\rangle \notin \mathcal{H}_N$

Summary so far

- Particle number broken: $|\Psi\rangle \notin \mathcal{H}_N$
- Symmetry breaking encoded in anomalous density matrix

$$\kappa_{\mu\nu} = \langle \Psi | a_\nu a_\mu | \Psi \rangle \neq 0.$$

Summary so far

- Particle number broken: $|\Psi\rangle \notin \mathcal{H}_N$
- Symmetry breaking encoded in anomalous density matrix

$$\kappa_{\mu\nu} = \langle \Psi | a_\nu a_\mu | \Psi \rangle \neq 0.$$

- Half of Fock space as variational space

$$|\Psi\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2 \oplus \ldots \qquad \textbf{or} \qquad |\Psi\rangle \in \mathcal{H}_1 \oplus \mathcal{H}_3 \oplus \ldots$$

Summary so far

- Particle number broken: $|\Psi\rangle \notin \mathcal{H}_N$
- Symmetry breaking encoded in anomalous density matrix

$$\kappa_{\mu\nu} = \langle \Psi | a_\nu a_\mu | \Psi \rangle \neq 0.$$

- Half of Fock space as variational space

$$|\Psi\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2 \oplus \ldots \qquad \textbf{or} \qquad |\Psi\rangle \in \mathcal{H}_1 \oplus \mathcal{H}_3 \oplus \ldots$$

- Larger variational space implies $E^{\mathrm{HFB}} \leq E^{\mathrm{HF}}$

Summary so far

- Particle number broken: $|\Psi\rangle \notin \mathcal{H}_N$
- Symmetry breaking encoded in anomalous density matrix

$$\kappa_{\mu\nu} = \langle \Psi | a_\nu a_\mu | \Psi \rangle \neq 0.$$

- Half of Fock space as variational space

$$|\Psi\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2 \oplus \ldots \qquad \textbf{or} \qquad |\Psi\rangle \in \mathcal{H}_1 \oplus \mathcal{H}_3 \oplus \ldots$$

- Larger variational space implies $E^{\mathrm{HFB}} \leq E^{\mathrm{HF}}$
- Bogoliubov states are characterised (up to a phase) by their **idempotent** generalized density matrix

$$\mathcal{R} = \begin{pmatrix} \rho & \kappa \\ -\kappa^* & 1 - \rho^* \end{pmatrix}$$

Equations of motion

Minimize $E[\rho, \kappa, \kappa^*]$ subject to 1. $\mathcal{R}^2 = \mathcal{R}$ and 2. $\langle \Psi | N | \Psi \rangle = N_0$.

Equations of motion

Minimize $E[\rho, \kappa, \kappa^*]$ subject to 1. $\mathcal{R}^2 = \mathcal{R}$ and 2. $\langle \Psi | N | \Psi \rangle = N_0$.

- Set $\delta \Big[E[\rho, \kappa, \kappa^*] - \lambda \, \mathrm{Tr}\{\rho\} - \mathrm{Tr}\{\Lambda(\mathcal{R}^2 - \mathcal{R})\} \Big] = 0$

Equations of motion

Minimize $E[\rho, \kappa, \kappa^*]$ subject to 1. $\mathcal{R}^2 = \mathcal{R}$ and 2. $\langle \Psi | N | \Psi \rangle = N_0$.

- Set $\delta \Big[E[\rho, \kappa, \kappa^*] - \lambda \, \mathrm{Tr}\{\rho\} - \mathrm{Tr}\{\Lambda(\mathcal{R}^2 - \mathcal{R})\} \Big] = 0$
- this leads to

Fermi energy (now uniquely defined)

$$[\mathcal{H}, \mathcal{R}] = 0 \implies \mathcal{H} \begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix} \equiv \begin{pmatrix} h - \lambda & \Delta \\ -\Delta^* & -h^* + \lambda \end{pmatrix} \begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix} = E_\mu \begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix}$$

$$h_{\alpha\gamma} \equiv \frac{\delta E[\rho, \kappa, \kappa^*]}{\delta \rho_{\gamma\alpha}} = t_{\alpha\gamma} + \sum_{\beta\delta} \bar{v}_{\alpha\beta\gamma\delta} \, \rho_{\delta\beta} = \text{Hartree–Fock potential}$$

$$\Delta_{\alpha\gamma} \equiv \frac{\delta E[\rho, \kappa, \kappa^*]}{\delta \kappa^*_{\alpha\gamma}} = \frac{1}{2} \sum_{\beta\delta} \bar{v}_{\alpha\gamma\beta\delta} \, \kappa_{\beta\delta} = \text{Pairing potential}$$

Equations of motion: discussion

$$\mathcal{H}\begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix} = \begin{pmatrix} h-\lambda & \Delta \\ -\Delta^* & -h^*+\lambda \end{pmatrix}\begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix} = E_\mu \begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix}$$

- $\{(U,V)_\mu\}$ and $\{E_\mu\}$ denote quasi-particle states and energies
- The field h generates shell structure while Δ drives pair scattering
- Not a standard eigenvalue problem as $\mathcal{H} = \mathcal{H}[\rho,\kappa,\kappa^*] = \mathcal{H}[U,V]$
- In a s.p. space of dimension M, $\mathcal{H}$ has $2M$ eigenvalues.

1. They come in pairs: $(U_\mu, V_\mu) \to E_\mu$ then $(V^*_\mu, U^*_\mu) \to -E_\mu$. Check this.
2. One has to choose **only one** of each pair to construct $|\Psi\rangle$
3. ...simplest: choose $E_\mu \geq 0$ to get state with lowest energy.

Iterative procedure

In

1. H
2. N
3. $|i\rangle^{(0)}$
4. $U^{(0)}, V^{(0)}$

$\longrightarrow$

$$U^{(n+1)}, V^{(n+1)}$$

$$\rho^{(n)}, \kappa^{(n)} \quad \mathrm{HFB} \quad h^{(n)}, \Delta^{(n)}$$

$$E[\rho, \kappa, \kappa^*]^{(n)}$$

$\longrightarrow$

Out

1. $E^{(\mathrm{HFB})}$
2. ρ, κ
3. E_μ
4. ...

Iterative procedure

In

1. H
2. N
3. $|i\rangle^{(0)}$
4. $U^{(0)}, V^{(0)}$

$\longrightarrow$

$U^{(n+1)}, V^{(n+1)}$

$\rho^{(n)}, \kappa^{(n)}$ HFB $h^{(n)}, \Delta^{(n)}$

$E[\rho, \kappa, \kappa^*]^{(n)}$

$\longrightarrow$

Out

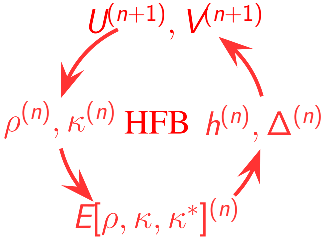
1. $E^{(\mathrm{HFB})}$
2. ρ, κ
3. E_μ
4. ...

- Not shown:
  - determination of $\lambda^{(n+1)}$ as a function of $h^{(n)}, \Delta^{(n)}$
  - choice of M among $2M$ possible $U^{(n+1)}, V^{(n+1)}$

Iterative procedure

In

1. H
2. N
3. $|i\rangle^{(0)}$
4. $U^{(0)}, V^{(0)}$



Out

1. $E^{(\mathrm{HFB})}$
2. ρ, κ
3. E_μ
4. ...

- Not shown:
  - determination of $\lambda^{(n+1)}$ as a function of $h^{(n)}, \Delta^{(n)}$
  - choice of M among $2M$ possible $U^{(n+1)}, V^{(n+1)}$
- The end result is the energy-optimal Bogoliubov state $|\Psi\rangle$

Iterative procedure

In

1. H
2. N
3. $|i\rangle^{(0)}$
4. $U^{(0)}, V^{(0)}$

$\longrightarrow$

$U^{(n+1)}, V^{(n+1)}$

$\rho^{(n)}, \kappa^{(n)}$ HFB $h^{(n)}, \Delta^{(n)}$

$E[\rho, \kappa, \kappa^*]^{(n)}$

$\longrightarrow$

Out

1. $E^{(\mathrm{HFB})}$
2. ρ, κ
3. E_μ
4. ...

- Not shown:
  - determination of $\lambda^{(n+1)}$ as a function of $h^{(n)}, \Delta^{(n)}$
  - choice of M among $2M$ possible $U^{(n+1)}, V^{(n+1)}$
- The end result is the energy-optimal Bogoliubov state $|\Psi\rangle$
- Potentials (h, Δ) and (U, V) will depend on N in non-trivial way

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

where $(\bar{U}, \bar{V})$ have a very particular form

$$\bar{U} = \begin{pmatrix} 0 & \cdots & & & & & & \cdots & & 0 \\ \vdots & \ddots & & & & & & & & \vdots \\ & & u_1 & 0 & & & & & & \\ & & 0 & u_1 & & & & & & \\ & & & & \ddots & & & & & \\ & & & & & u_n & 0 & & & \\ & & & & & 0 & u_n & & & \\ & & & & & & & 1 & & \\ & & & & & & & & \ddots & \\ 0 & & & & & & & & & 1 \end{pmatrix}$$

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

where $(\bar{U}, \bar{V})$ have a very particular form

$$\bar{V} = \begin{pmatrix}
1 & 0 & \dots & & & & & & \dots & 0 \\
0 & \ddots & & & & & & & & \vdots \\
\vdots & & 0 & v_1 & & & & & & \\
 & & -v_1 & 0 & & & & & & \\
 & & & & \ddots & & & & & \\
 & & & & & 0 & v_n & & & \\
 & & & & & -v_n & 0 & & & \\
 & & & & & & & 0 & & \\
 & & & & & & & & \ddots & \\
0 & & & & & & & & & 0
\end{pmatrix}$$

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

Three successive transformations

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

Three successive transformations

1. Canonical single-particle basis $\{b, b^\dagger\}$ $\rightarrow$ diagonalises ρ

$$\begin{pmatrix} b \\ b^\dagger \end{pmatrix} = \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}$$

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

Three successive transformations

1. Canonical single-particle basis $\{b, b^\dagger\}$ $\rightarrow$ diagonalises ρ

$$\begin{pmatrix} b \\ b^\dagger \end{pmatrix} = \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}$$

2. Canonical quasi-particle basis $\{\alpha, \alpha^\dagger\}$ $\rightarrow$ see next slide

$$\begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix} = \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} b \\ b^\dagger \end{pmatrix}$$

Bloch-Messiah-Zumino decomposition

$$\begin{pmatrix} U^\dagger & V^\dagger \\ V^T & U^T \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix}$$

Three successive transformations

1. Canonical single-particle basis $\{b, b^\dagger\}$ $\rightarrow$ diagonalises ρ

$$\begin{pmatrix} b \\ b^\dagger \end{pmatrix} = \begin{pmatrix} C^\dagger & 0 \\ 0 & C^T \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}$$

2. Canonical quasi-particle basis $\{\alpha, \alpha^\dagger\}$ $\rightarrow$ see next slide

$$\begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix} = \begin{pmatrix} \bar{U}^\dagger & \bar{V}^\dagger \\ \bar{V}^T & \bar{U}^T \end{pmatrix} \begin{pmatrix} b \\ b^\dagger \end{pmatrix}$$

3. Bogoliubov quasi-particle basis $\{\beta, \beta^\dagger\}$ $\rightarrow$ does not affect observables

$$\begin{pmatrix} \beta \\ \beta^\dagger \end{pmatrix} = \begin{pmatrix} D^\dagger & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix}$$

The canonical quasiparticle basis

- The particular form of $\bar{U}$ and $\bar{V}$ defines **canonically conjugate** states $(\mu, \bar{\mu})$

$$U \sim \begin{pmatrix} u_\mu & 0 \\ 0 & u_\mu \end{pmatrix}, \qquad V \sim \begin{pmatrix} 0 & v_\mu \\ -v_\mu & 0 \end{pmatrix}$$

The canonical quasiparticle basis

- The particular form of $\bar{U}$ and $\bar{V}$ defines **canonically conjugate** states $(\mu, \bar{\mu})$

$$U \sim \begin{pmatrix} u_\mu & 0 \\ 0 & u_\mu \end{pmatrix}, \qquad V \sim \begin{pmatrix} 0 & v_\mu \\ -v_\mu & 0 \end{pmatrix}$$

- ...which induce substructure in the generalized density matrix

$$\rho \sim \begin{pmatrix} v_\mu^2 & 0 \\ 0 & v_\mu^2 \end{pmatrix} \qquad \kappa \sim \begin{pmatrix} 0 & +u_\mu v_\mu \\ -u_\mu v_\mu & 0 \end{pmatrix}$$

The canonical quasiparticle basis

- The particular form of $\bar{U}$ and $\bar{V}$ defines **canonically conjugate** states $(\mu, \bar{\mu})$

$$U \sim \begin{pmatrix} u_\mu & 0 \\ 0 & u_\mu \end{pmatrix} , \qquad V \sim \begin{pmatrix} 0 & v_\mu \\ -v_\mu & 0 \end{pmatrix}$$

- ...which induce substructure in the generalized density matrix

$$\rho \sim \begin{pmatrix} v_\mu^2 & 0 \\ 0 & v_\mu^2 \end{pmatrix} \qquad \kappa \sim \begin{pmatrix} 0 & +u_\mu v_\mu \\ -u_\mu v_\mu & 0 \end{pmatrix}$$

- . . . so the operators $\alpha, \alpha^\dagger$ are particularly simple

$$\alpha_\mu^\dagger = u_\mu b_\mu^\dagger - v_\mu b_{\bar{\mu}} \, ,$$
$$\alpha_\mu = u_\mu b_\mu - v_\mu b_{\bar{\mu}}^\dagger \, ,$$
$$\text{with } u_\mu^2 + v_\mu^2 = 1 \, .$$

Structure of the Bogoliubov state

Three different kinds of s.p. states

- Occupied: $v_\mu = 1, u_\mu = 0, \rho_{\mu\mu} = 1, \alpha^\dagger_\mu = b_\mu$

Structure of the Bogoliubov state

Three different kinds of s.p. states

- Occupied: $v_\mu = 1, u_\mu = 0, \rho_{\mu\mu} = 1, \alpha^\dagger_\mu = b_\mu$
- Paired: : $0 < v_\mu < 1, 0 < u_\mu < 1, \alpha^\dagger_\mu = u_\mu b^\dagger_\mu - v_\mu b_{\bar{\mu}}$

Structure of the Bogoliubov state

Three different kinds of s.p. states

- Occupied: $v_\mu = 1, u_\mu = 0, \rho_{\mu\mu} = 1, \alpha^\dagger_\mu = b_\mu$
- Paired: : $0 < v_\mu < 1, 0 < u_\mu < 1, \alpha^\dagger_\mu = u_\mu b^\dagger_\mu - v_\mu b_{\bar{\mu}}$
- Empty: $u_\mu = 1, v_\mu = 0, \rho_{\mu\mu} = 0, \alpha^\dagger_\mu = b^\dagger_\mu$

Structure of the Bogoliubov state

Three different kinds of s.p. states

- Occupied: $v_\mu = 1, u_\mu = 0, \rho_{\mu\mu} = 1, \alpha^\dagger_\mu = b_\mu$
- Paired: : $0 < v_\mu < 1, 0 < u_\mu < 1, \alpha^\dagger_\mu = u_\mu b^\dagger_\mu - v_\mu b_{\bar{\mu}}$
- Empty: $u_\mu = 1, v_\mu = 0, \rho_{\mu\mu} = 0, \alpha^\dagger_\mu = b^\dagger_\mu$

runs over occupied states

runs over half of the paired states

$$|\Psi\rangle = \prod_i b^\dagger_i \prod_{\mu>0} \left(u_\mu + v_\mu b^\dagger_\mu b^\dagger_{\bar{\mu}} \right) |0\rangle \, .$$

Bardeen-Cooper-Schrieffer (BCS) structure

Structure of the Bogoliubov state

Three different kinds of s.p. states

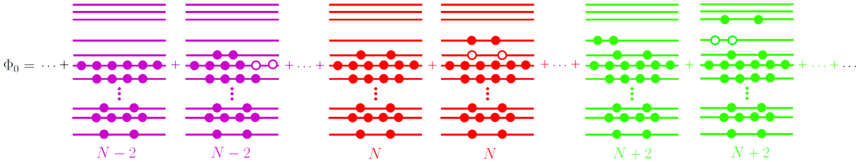
- Occupied: $v_\mu = 1, u_\mu = 0, \rho_{\mu\mu} = 1, \alpha_\mu^\dagger = b_\mu$
- Paired: : $0 < v_\mu < 1, 0 < u_\mu < 1, \alpha_\mu^\dagger = u_\mu b_\mu^\dagger - v_\mu b_{\bar{\mu}}$
- Empty: $u_\mu = 1, v_\mu = 0, \rho_{\mu\mu} = 0, \alpha_\mu^\dagger = b_\mu^\dagger$

runs over occupied states

runs over half of the paired states

$$|\Psi\rangle = \prod_i b_i^\dagger \prod_{\mu>0} \left(u_\mu + v_\mu b_\mu^\dagger b_{\bar{\mu}}^\dagger\right) |0\rangle \, .$$

Bardeen-Cooper-Schrieffer (BCS) structure



Equations in the canonical basis

Specific structure of $\bar{U}, \bar{V}$ leads to

$$\begin{pmatrix} u_\mu^2 \\ v_\mu^2 \end{pmatrix} = \frac{1}{2}\left(1 \pm \frac{\bar{h}_\mu}{\sqrt{\bar{h}_\mu^2 + \Delta_{\mu\bar{\mu}}^2}}\right) ,$$

$$\bar{h}_\mu = \frac{1}{2}\left[h_{\mu\mu} + h_{\bar{\mu}\bar{\mu}} - 2\lambda\right] .$$

...which leads to

$$\rho_{\mu\nu} = v_\mu^2 \delta_{\mu\nu} ,$$

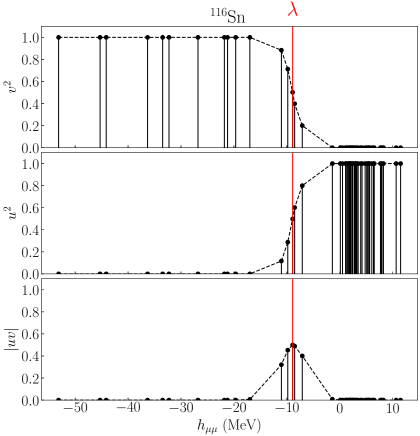
$$\kappa_{\mu\bar{\nu}} = u_\mu v_\mu .$$

Equations in the canonical basis

Specific structure of $\bar{U}$, $\bar{V}$ leads to

$$\begin{pmatrix} u_\mu^2 \\ v_\mu^2 \end{pmatrix} = \frac{1}{2}\left(1 \pm \frac{\bar{h}_\mu}{\sqrt{\bar{h}_\mu^2 + \Delta_{\mu\bar{\mu}}^2}}\right) ,$$

$$\bar{h}_\mu = \frac{1}{2}\left[h_{\mu\mu} + h_{\bar{\mu}\bar{\mu}} - 2\lambda\right] .$$



...which leads to

$$\rho_{\mu\nu} = v_\mu^2 \delta_{\mu\nu} ,$$

$$\kappa_{\mu\bar{\nu}} = u_\mu v_\mu .$$

The quasiparticle spectrum

In the canonical basis

$$E_\mu \approx \sqrt{\bar{h}_\mu^2 + \Delta_{\mu\bar{\mu}}^2}\,.$$

The quasiparticle spectrum

In the canonical basis

$$E_\mu \approx \sqrt{\bar{h}_\mu^2 + \Delta^2_{\mu\bar{\mu}}}\,.$$

Slater determinant

- Lowest-energy excitation

$$|\Phi_i^a\rangle \equiv b_a^\dagger b_i |\Phi\rangle$$

- Spectrum

$$E_{ai}^{(\mathrm{HF})} - E_0^{(\mathrm{HF})} \approx h_{aa} - h_{ii} \geq 0$$

The quasiparticle spectrum

In the canonical basis

$$E_\mu \approx \sqrt{\bar{h}_\mu^2 + \Delta_{\mu\bar{\mu}}^2}\,.$$

Slater determinant

- Lowest-energy excitation

$$|\Phi_i^a\rangle \equiv b_a^\dagger b_i |\Phi\rangle$$

- Spectrum

$$E_{ai}^{(\mathrm{HF})} - E_0^{(\mathrm{HF})} \approx h_{aa} - h_{ii} \geq 0$$

Bogoliubov state

- Lowest-energy excitation

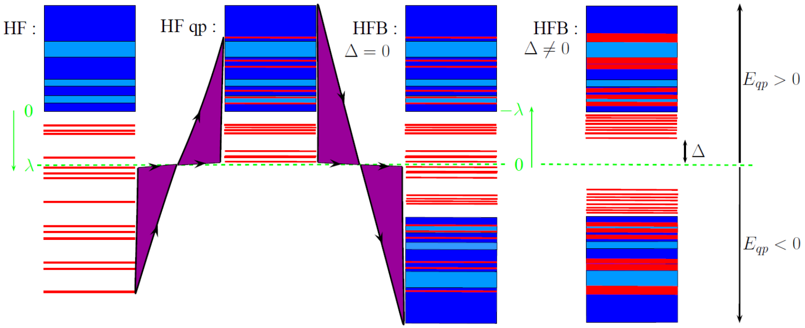
$$|\Phi^{\mu\nu}\rangle \equiv \beta_\mu^\dagger \beta_\nu^\dagger |\Phi\rangle$$

- Spectrum

$$E_{\mu\nu}^{(\mathrm{HFB})} - E_0^{(\mathrm{HFB})} \approx E_\mu + E_\nu \geq 2\,\Delta_\lambda$$

Δ for s.p. states near the Fermi energy

The quasiparticle spectrum



The Bardeen-Cooper-Schrieffer (BCS) limit

- Assumptions[1]:
  - the canonical basis is identical to the Hartree-Fock basis
  - canonically paired states $(\alpha, \bar{\alpha})$ are linked through time-reversal

$$
\begin{aligned}
h_{\alpha\beta} = h_{\bar{\alpha}\bar{\beta}} &= \epsilon_\alpha \delta_{\alpha\beta} & \qquad \rho_{\alpha\beta} = \rho_{\bar{\alpha}\bar{\beta}} &= v_\alpha^2 \delta_{\alpha\beta} \, , \\
\kappa_{\alpha\bar{\beta}} &= u_\alpha v_\alpha \delta_{\alpha\beta} & \qquad \kappa_{\bar{\alpha}\beta} &= -u_\alpha v_\alpha \delta_{\alpha\beta}
\end{aligned}
$$

[1] Typically okay for ground states of well-bound even-even nuclei

The Bardeen-Cooper-Schrieffer (BCS) limit

- Assumptions[1]:
  - the canonical basis is identical to the Hartree-Fock basis
  - canonically paired states $(\alpha, \bar{\alpha})$ are linked through time-reversal

$$
\begin{aligned}
h_{\alpha\beta} = h_{\bar{\alpha}\bar{\beta}} &= \epsilon_\alpha \delta_{\alpha\beta} \qquad & \rho_{\alpha\beta} = \rho_{\bar{\alpha}\bar{\beta}} &= v_\alpha^2 \delta_{\alpha\beta} \,, \\
\kappa_{\alpha\bar{\beta}} &= u_\alpha v_\alpha \delta_{\alpha\beta} \qquad & \kappa_{\bar{\alpha}\beta} &= -u_\alpha v_\alpha \delta_{\alpha\beta}
\end{aligned}
$$

- Structure of ρ and κ in the canonical basis (see slide 24 in Lecture 5)

$$
\rho \sim \begin{pmatrix} v_\mu^2 & 0 \\ 0 & v_\mu^2 \end{pmatrix} \qquad \kappa \sim \begin{pmatrix} 0 & +u_\mu v_\mu \\ -u_\mu v_\mu & 0 \end{pmatrix}
$$

[1] Typically okay for ground states of well-bound even-even nuclei

Blocking

- Picking $E_\mu \geq 0$ results in $|\Psi\rangle$ with even number parity.[2]

[2] Most of the time, though not always.

Blocking

- Picking $E_\mu \geq 0$ results in $|\Psi\rangle$ with even number parity.[2]
- Model systems with odd number of particles through qp excitation

$$|\Phi^\mu\rangle \equiv \boxed{\beta^\dagger_\mu} |\Phi\rangle \text{ has the opposite number parity of } |\Phi\rangle$$

Choice necessary and not obvious a priori

[2]Most of the time, though not always.

Blocking

- Picking $E_\mu \geq 0$ results in $|\Psi\rangle$ with even number parity.[2]
- Model systems with odd number of particles through qp excitation

$$|\Phi^\mu\rangle \equiv \boxed{\beta^\dagger_\mu} |\Phi\rangle \text{ has the opposite number parity of } |\Phi\rangle$$

Choice necessary and not obvious a priori

- Equivalent to picking $-E_\mu$ as opposed to E_μ when constructing $|\Phi\rangle$

[2] Most of the time, though not always.

Blocking

- Picking $E_\mu \geq 0$ results in $|\Psi\rangle$ with even number parity.[2]
- Model systems with odd number of particles through qp excitation

$$|\Phi^\mu\rangle \equiv \boxed{\beta^\dagger_\mu}\,|\Phi\rangle \text{ has the opposite number parity of } |\Phi\rangle$$

Choice necessary and not obvious a priori

- Equivalent to picking $-E_\mu$ as opposed to E_μ when constructing $|\Phi\rangle$
- This **breaks a pair**.
  - one set of canonical partners $(\kappa, \bar{\kappa})$ with

$$\rho_{\kappa\kappa} = 1 \text{ but } \rho_{\bar{\kappa}\bar{\kappa}} = 0 \text{ and } \kappa_{\kappa,\bar{\kappa}} = 0\,.$$

  - structure of $|\Phi^\mu\rangle$

$$|\Phi^\mu\rangle = a^\dagger_\kappa \prod_{\mu>0,\mu\neq\kappa} (u_\mu + v_\mu b^\dagger_\mu b^\dagger_{\bar{\mu}})|0\rangle\,.$$

[2] Most of the time, though not always.

Blocking

κ and $\bar{\kappa}$ are **blocked** from participating in pairing

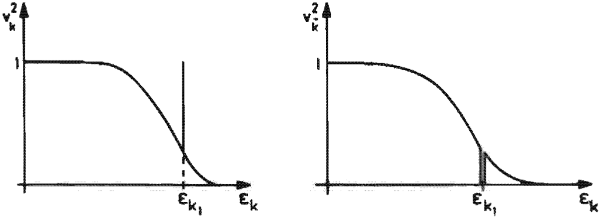


Figure 6.7. Blocking of the state k_1.

Fig. from. Ring & Schuck, Springer Verlag (Berlin), 1986.

Consequences of blocking

- Blocking effect reduces pairing energy

$$|E_{\text{pair}}(|\Phi^{\mu}\rangle)| < |E_{\text{pair}}(|\Phi\rangle)| = \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\kappa^{*}_{\alpha\beta}\kappa_{\gamma\delta}\,.$$

Consequences of blocking

- Blocking effect reduces pairing energy

$$|E_{\text{pair}}(|\Phi^{\mu}\rangle)| < |E_{\text{pair}}(|\Phi\rangle)| = \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta}\kappa^{*}_{\alpha\beta}\kappa_{\gamma\delta}\,.$$

- Lowest excitation energy smaller

$$\langle\Phi^{\nu}|H|\Phi^{\nu}\rangle - \langle\Phi^{\mu}|H|\Phi^{\mu}\rangle = E_{\nu} - E_{\mu} \ngeq 2\Delta_{\lambda}$$

Consequences of blocking

- Blocking effect reduces pairing energy

$$|E_{\text{pair}}(|\Phi^{\mu}\rangle)| < |E_{\text{pair}}(|\Phi\rangle)| = \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} \kappa^{*}_{\alpha\beta} \kappa_{\gamma\delta} \, .$$

- Lowest excitation energy smaller

$$\langle \Phi^{\nu} | H | \Phi^{\nu} \rangle - \langle \Phi^{\mu} | H | \Phi^{\mu} \rangle = E_{\nu} - E_{\mu} \not\geq 2\Delta_{\lambda}$$

- Much harder to model in practice
  1. Small energy differences between different states
  2. Choice of quasiparticle index at every iteration → Convergence issues

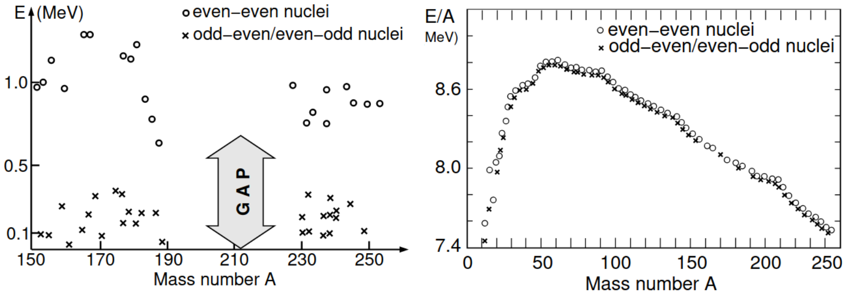


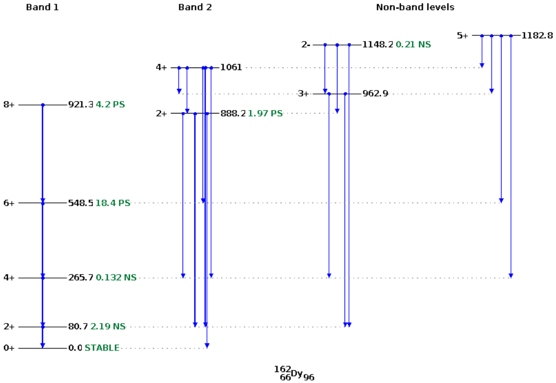
Figure of D. Page et al. arxiv:1302.6626 (2013).

Signs of pairing noticed early on

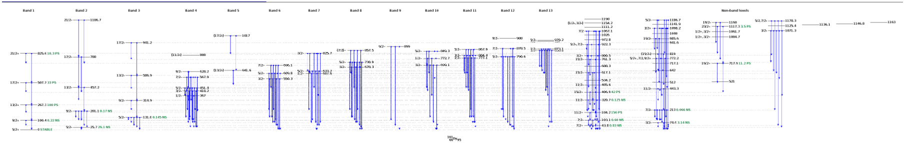
Bohr et al., Phys. Rev. 110 936 (1958)

1. Spectral gap for even-even nuclei
2. Binding energy penalty for non-even-even systems

Level density of ^{162}Dy



Level density of ^{161}Dy



Comparison

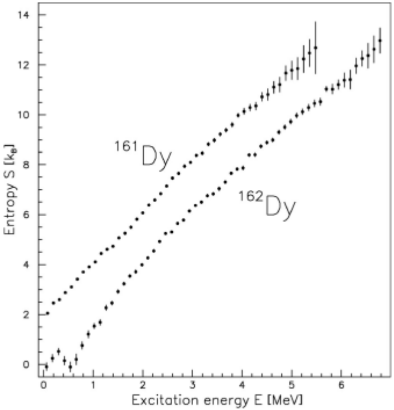


FIG. 26. Observed entropy for ^{161,162}Dy as a function of excitation energy E. From Guttormsen *et al.*, 2000.

Entropy(E) $\sim$ log(number of states at given excitation energy).

Energy staggering

- For even N, a perturbative treatment gives:

$$
\begin{aligned}
E^{\mathrm{HFB}}(N \pm 1) &\approx E^{\mathrm{HFB}}(N) \mp \lambda + \min_{\mu} \left(E_{\mu} \right) \\
&\approx E^{\mathrm{HFB}}(N) \mp \lambda + \Delta_{\lambda}
\end{aligned}
$$

Energy staggering

- For even N, a perturbative treatment gives:

$$\begin{aligned} E^{\mathrm{HFB}}(N \pm 1) &\approx E^{\mathrm{HFB}}(N) \mp \lambda + \min_{\mu} (E_\mu) \\ &\approx E^{\mathrm{HFB}}(N) \mp \lambda + \Delta_\lambda \end{aligned}$$

- $$\Delta_\lambda \approx \frac{(-1)^{N+1}}{2} \left[E^{\mathrm{HFB}}(N+1) + E^{\mathrm{HFB}}(N-1) - 2E^{\mathrm{HFB}}(N) \right].$$

Energy staggering

- For even N, a perturbative treatment gives:

$$
\begin{aligned}
E^{\mathrm{HFB}}(N \pm 1) &\approx E^{\mathrm{HFB}}(N) \mp \lambda + \min_{\mu}\left(E_{\mu}\right) \\
&\approx E^{\mathrm{HFB}}(N) \mp \lambda + \Delta_{\lambda}
\end{aligned}
$$

-

$$
\Delta_{\lambda} \approx \frac{(-1)^{N+1}}{2}\left[E^{\mathrm{HFB}}(N+1) + E^{\mathrm{HFB}}(N-1) - 2E^{\mathrm{HFB}}(N)\right].
$$

- which can be compared to experiment

$$
\Delta_n^{(3)}(N, Z) \equiv \frac{(-1)^{N+1}}{2}\left[BE(Z, N+1) + BE(Z, N-1) - 2BE(Z, N)\right].
$$

Energy staggering

- For even N, a perturbative treatment gives:

$$\begin{aligned} E^{\mathrm{HFB}}(N \pm 1) &\approx E^{\mathrm{HFB}}(N) \mp \lambda + \min_{\mu}\left(E_{\mu}\right) \\ &\approx E^{\mathrm{HFB}}(N) \mp \lambda + \Delta_{\lambda} \end{aligned}$$

-

$$\Delta_{\lambda} \approx \frac{(-1)^{N+1}}{2}\left[E^{\mathrm{HFB}}(N+1) + E^{\mathrm{HFB}}(N-1) - 2E^{\mathrm{HFB}}(N)\right].$$

- which can be compared to experiment

$$\Delta_n^{(3)}(N,Z) \equiv \frac{(-1)^{N+1}}{2}\left[BE(Z,N+1) + BE(Z,N-1) - 2BE(Z,N)\right].$$

- Models typically tune their pairing interaction by fitting Δ_{λ} to $\Delta^{(3)}$

Energy staggering

- For even N, a perturbative treatment gives:

$$E^{\mathrm{HFB}}(N \pm 1) \approx E^{\mathrm{HFB}}(N) \mp \lambda + \min_{\mu} (E_\mu)$$
$$\approx E^{\mathrm{HFB}}(N) \mp \lambda + \Delta_\lambda$$

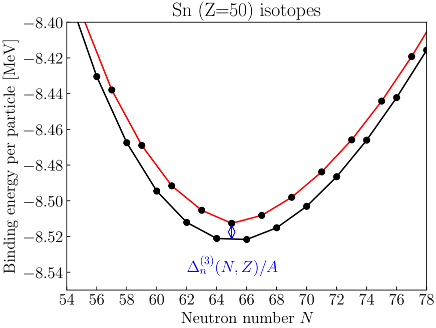
- $$\Delta_\lambda \approx \frac{(-1)^{N+1}}{2} \left[E^{\mathrm{HFB}}(N+1) + E^{\mathrm{HFB}}(N-1) - 2E^{\mathrm{HFB}}(N)\right].$$

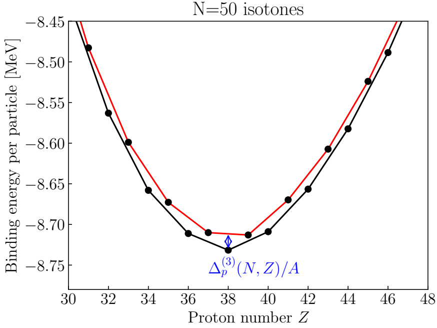
- which can be compared to experiment

$$\Delta_n^{(3)}(N,Z) \equiv \frac{(-1)^{N+1}}{2} \left[BE(Z,N+1) + BE(Z,N-1) - 2BE(Z,N)\right].$$

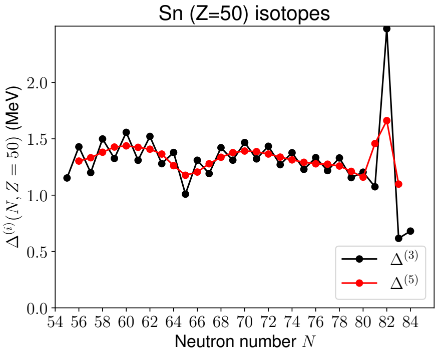
- Models typically tune their pairing interaction by fitting Δ_λ to $\Delta^{(3)}$
- ...even if this perturbative argument misses tons of physics!

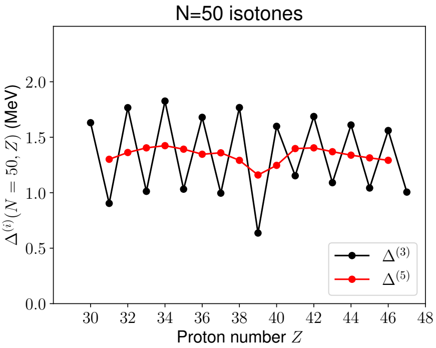
Odd-even mass staggering





Odd-even mass staggering





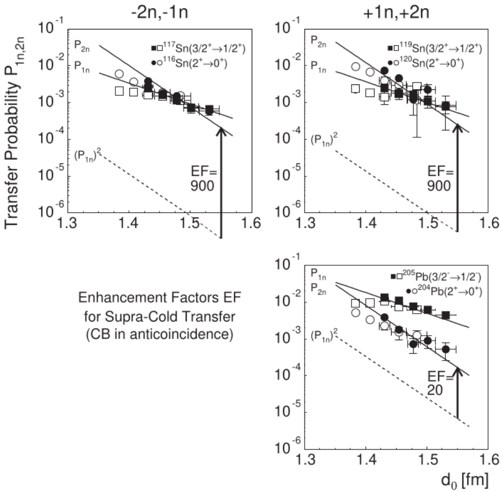
- Higher order differences smooth things out

$$\Delta_n^{(5)}(N, Z) \equiv \frac{(-1)^N}{8}\Big[BE(Z, N+2) - 4BE(Z, N+1) + 6BE(Z, N) - 4BE(Z, N-1) + E(N-2)\Big].$$

- ...but s.p. structure remains visible!

Transfer reactions

$$P(\text{transfer of two nucleons}) >> [P(\text{transfer of one nucleon})]^2$$



The end.